

## Warm-up

Compute the following integrals:

$$(1) \int (x - 3) \sin(x) \, dx, \quad (2) \int x \ln(x) \, dx.$$

## Medium

Compute the following integrals:

$$(1) \int x^2 e^{2x} \, dx, \quad (2) \int \cos^3(x) \sin^4(x) \, dx, \quad (3) \int e^x \sin(5x) \, dx,$$
$$(4) \int \arctan(x) \, dx, \quad (5) \int \sec^4(x) \tan^3(x) \, dx.$$

## Stretch

(1) Compute  $\int \sec^3(x) \, dx$  using integration by parts.

(2) Use integration by parts and the Pythagorean theorem  $\sin^2(x) + \cos^2(x) = 1$  to derive the following *reduction formula* for powers of  $\sin(x)$ :

$$\int \sin^n(x) \, dx = -\frac{1}{n} \sin^{n-1}(x) \cos(x) + \frac{n-1}{n} \int \sin^{n-2}(x) \, dx.$$